NUHA

UI and UX Style Research for the New Version

Colour • Arabic type • Illustration • Interaction • Motion and sound • Rewards

Client research brief | 20 October 2026

Keep NUHA calm, stable, and adult-tunable

A desk-research synthesis of what visual and interaction style the new Kotlin app should adopt for Arabic-speaking children with autism, ADHD, and learning difficulties, and how that style maps onto the existing Jetpack Compose code.

How to read this report

Each recommendation carries an evidence level: Strong, Moderate, Guideline, or Inference. Inferences are design judgements that must be tested with children on the real table.

No child sessions or user tests were conducted for this report. No medical benefit is asserted.

5 research tracks • 14 style recommendations • 9 items to test with children

Summary

NUHA's new version should keep its calm, low-saturation teal/sand direction, but treat it as a sensible default rather than a proven teaching tool. Around that default it needs three things the current skeleton lacks: verified luminance contrast, a bundled and tested Arabic type system, and a settings layer per child that adults can use to raise or lower sensory load. The best-supported findings in this research do not concern colour. They concern structure. Irrelevant decoration lowers young children's learning. Multimedia that is congruent with the content helps, while interactive extras distract. Autistic children vary widely in their sensory profiles, so no single palette or soundscape suits everyone. Visual schedules, prompting with fading, and immediate, consistent feedback are established practice in autism support. Taken together, the evidence points to a specific style. Screens are sparse and predictable, with one task, oversized targets that never move, and tap-first input. A single quiet guide character speaks in Arabic and demonstrates each task. Errors get a neutral re-prompt instead of a buzzer. A small token board fills up during the session. Motion is brief and only ever points at the learning target. Music is off by default, which suits both sensory needs and Saudi family norms. Much remains unproven: no controlled study compares muted and saturated app palettes for this population, no study tests Arabic typefaces with pre-readers, and the numeral convention in current Saudi primary textbooks is still unverified. Several of the recommendations below are therefore hypotheses to test with children on the real table, and they are labelled as such.

Evidence levels used in this report. Strong means a meta-analysis or several consistent studies, usually with typically developing or adjacent populations. **Moderate** means one or a few empirical studies, small samples, or indirect populations. **Guideline** means a standard, platform rule, or expert consensus without controlled trials. **Inference** means a design judgement derived from the above, and it must be tested.

Muted colour is harm reduction; luminance contrast does the real work

The direct evidence on autism and colour is thin, but it points one way. In a study of 29 autistic boys aged 4-17, they were significantly less likely than typically developing peers to prefer yellow, and more likely to prefer green and brown. The authors attribute this to hypersensitivity. The study is still unreplicated, and it tested saturated primaries, not muted sand (Grandgeorge & Masataka 2016). A 2025 eye-tracking study of 34 autistic children aged 6-16 found **about 29% less total fixation time**, and **highly saturated images amplified the group differences** (J Autism Dev Disord 2025). That shows saturation changes how autistic children look at images. It does not show discomfort or worse learning. Guidelines say the same thing with more confidence than the data support. The GOV.UK autism poster advises "use simple colours" and warns against "bright contrasting colours," but its authors frame the advice as non-prescriptive and say to test with users wherever it conflicts with low-vision advice (GOV.UK). The honest conclusion is that NUHA's palette is defensible as comfort and harm reduction. No study shows that it improves learning.

A second finding complicates "muted" more than most design write-ups admit. Children with high-functioning autism showed **reduced chromatic discrimination**: on a hue test they performed like typically developing children about three years younger (Franklin et al. 2010). They were also less accurate at detecting coloured targets on coloured backgrounds (Franklin et al. 2008). In a palette where everything is a soft pastel, subtle hue differences are exactly what some of these children struggle to see. The answer is to separate two properties that are often blurred together. Saturation is what calms. Luminance contrast is what makes things legible. WCAG 2.2 sets the floor: **4.5:1 for normal text, 3:1 for large text, and 3:1 for UI component boundaries and meaningful graphics** (W3C 1.4.3; W3C 1.4.11). Colour must never be the only cue (W3C 1.4.1).

Running the current Theme.kt values through the WCAG formula shows where NUHA stands. Text passes comfortably. Body text on the off-white background is 14.6:1, primary teal on background is 5.4:1, and white on primary is 5.8:1. The tinted fills fail as boundaries. The primary container is only 1.22:1 against the background, the secondary (sand) container 1.18:1, surface variant 1.15:1, and white cards 1.07:1. The outline colour sits at exactly 3.00:1. So a sand or pale-teal answer tile with no border is, by WCAG's measure, close to invisible as a control. The fix keeps the calm look: every tappable tile gets a 2-3dp border at or above 3:1, such as a darkened teal, and state changes such as "selected" use shape, border weight and an icon as well as hue.

The decoration evidence is stronger than the colour evidence. In a lab classroom, 24 kindergarteners scored **42% when taught in a heavily decorated room and 55% in a sparse room**, and they spent more time off task in the decorated room (APS). Critics argue that novelty, not irrelevance, may drive the effect (Imuta & Scarf 2014). A meta-analysis of 58 studies puts the cost of "seductive details" at **g = −0.33** (WSU). A meta-analysis of 43 studies with 2,147 young children found that congruent animation, music and sound helped story comprehension, while hotspots and games distracted, especially for children from less stimulating homes (Takacs, Swart & Bus 2015). These findings justify plain task backgrounds, task objects as the only high-contrast elements, and scene art that stays the same across sessions so it stops being novel. Finally, autistic children fall into **3-5 distinct sensory subtypes**, including hyper-responsive and sensory-seeking profiles. This holds in samples that span ages 1-21 and 2-12, so it applies to either age range NUHA might choose (PMC8507349; Autism CRC). A calm default is right because over-stimulating a hyper-responsive child is the costlier mistake. Adults still need a way to raise stimulation for a child who disengages.

Arabic type should be simplified Naskh, fully vowelized, and never letter-spaced

The only child-relevant Arabic legibility evidence favours simplified Naskh-style letterforms. In one study, 33 Moroccan Grade 3 pupils made fewer reading errors with Simplified Arabic (Inderscience). In another, children aged 9-12 read Simplified Arabic well on screen at 14-18pt, and found size 10 "not readable" (Abubaker & Lu 2012). Adult eye-tracking work adds that heavy ligature stacking makes text harder to read (Chahine 2012). No study compares Kufi or geometric sans faces with Naskh for pre-readers. A dyslexia study with 64 children with dyslexia and 64 without found no effect of letterform, and pointed instead to vowelization and diglossia (ResearchGate). NUHA should therefore not promise "dyslexia-friendly font" benefits.

For bundling, **Noto Naskh Arabic (SIL OFL 1.1)** is the safest default for vowelized learning text, because its marks are reportedly carefully positioned (Fontsource). Informal comparisons rate Cairo and Tajawal's diacritics only "adequate" (The Arabic Keyboard). **Baloo Bhaijaan 2** is a rounded, child-friendly OFL variable family (weights 400-800) and a plausible choice for headings and UI labels (Google Fonts). No independent test of harakat collisions exists for any of these fonts, so NUHA must render its own test strings on the real device. Those strings should include shadda with kasra, tanween, sukun on tall letters, and marks on lam-alef. Two rules come straight from W3C's Arabic layout requirements. Do not apply letter-spacing, because moving joined characters apart "creates undesirable results." Do not use tatweel for spacing, because it widens letters, which in NUHA's case are the very letters children are learning to recognize (W3C alreq). Increase word spacing and line height instead.

The evidence on diacritics is among the strongest in this review. Vowelization aided word recognition for both poor and skilled readers (Abu-Rabia 1997). Wrong vowelization hurt accuracy (Abu-Rabia 1998). Second graders rely on diacritics for decoding, and schools phase them out around Grade 4 (Frontiers 2020; J Psycholinguist Res 2026). So every learning word, letter and sentence in NUHA should carry **full, native-speaker-checked tashkeel** across the whole 2-10 range. UI labels can stay lightly vowelized, because pre-readers will navigate by icon and audio anyway. Full tashkeel needs vertical room. The current theme's body styles use about 1.6× line height, which is adequate, but displaySmall (40/52sp, 1.3×) and headlineMedium (32/44sp, 1.375×) are too tight for stacked marks. Learning text should sit at roughly 1.6-1.8×. Diglossia also matters: even fully vowelized text is harder when the standard form differs from the spoken one (Saiegh-Haddad 2003). That argues for Fusha text with familiar Saudi vocabulary, and possibly optional dialect voice prompts for the youngest children.

Numerals are the most consequential open question. W3C lists Saudi Arabia as a region that uses Arabic-Indic digits (٠١٢٣) (W3C alreq §6.1). The country is moving toward Western digits, though: the 2023 road-sign manual mandates them on new signs (Wikipedia). No source confirms which form current Ministry of Education primary maths books use. Until someone checks the books, NUHA should keep the numeral system as one app-level setting that all maths content reads. It should never mix the two systems on a screen, and it should format numbers explicitly rather than trusting the device locale. RTL mirroring follows established rules. Mirror anything that encodes sequence: back and next, progress, schedule strips, carousels. Do not mirror media controls, checkmarks, symmetric icons, or charts (Material; Firefox RTL guidelines). Letter-tracing paths are content, not layout, so they must never be mirrored. Number lines need a curriculum check before NUHA picks a direction.

One quiet guide character, with realism wherever learning must transfer

Pedagogical agents have a small positive effect on learning, **g ≈ 0.20** (Birmingham 2021). The effect is larger for K-12 learners than for adults, across 43 studies (Schroeder et al. 2013). Very little of this evidence covers preschoolers, and almost none covers autistic or ADHD children using apps (RWTH). The Takacs meta-analysis supplies the operating rule: a character helps when it carries the content and hurts when it is a side attraction. NUHA should have **one guide character across all nine activities**. It appears at instruction and feedback moments, demonstrates the task (traces the letter, taps 1-2-3), and stays still while the child works. The Transporters is a useful precedent. Predictable, rule-governed vehicle characters with real human faces helped 20 autistic children aged 4-7 reach typical-level emotion recognition, and the authors credit the characters' mechanical predictability (Golan et al. 2010; Baron-Cohen et al.). For Saudi families, an animal or neutral mascot sidesteps questions of dress and gender. If it is an animal, it should not be a dog shown as an indoor pet.

Illustration fidelity should change with the activity rather than follow one house style. Toddlers aged 15-18 months learned labels better from photographs and realistic drawings than from cartoons, and transferred them to real objects better (Ganea et al. 2008). For environment matching, especially if NUHA targets ages 2-6, objects should therefore be realistic or semi-realistic and drawn from Saudi homes and streets: the dallah, dates, majlis, palm trees, the souq. Flat, simplified style belongs to UI chrome. Emotion work needs a ladder. Autistic children process cartoon faces more typically than real ones (Rosset et al. 2008). But a meta-analysis of about 595 participants found that gains from emotion training were robust in the short term and not maintained at follow-up (Zhao et al.). The emotion activity should therefore start with simplified faces and progress to real photographs of diverse Saudi and Gulf children and adults, including girls and women in hijab and men in ghutra, under varied lighting. It should use several examples per emotion.

Cultural fit rests mostly on convention, not research. The notes found no Saudi regulator guidance written specifically for children's educational apps. Gmedia's age classification covers film and games (Gmedia), and a review notes a shortage of culturally adapted Arabic apps for autistic children (ScienceDirect). Market evidence is clear on one point: Islamic children's apps sell a no-music option. Kahf Kids, Miraj Stories and Durioo+ all offer it (Kahf Kids; Miraj). The remaining guidance is design judgement for Saudi educators to review. Characters should wear modest, locally recognizable dress. Family scenes should centre parents, grandparents and siblings. Quranic text and the name of Allah must never appear on anything a child taps, drags, erases or throws away.

Stable, oversized, tap-first controls are where the hardest evidence sits

Children miss small targets far more often than adults, and their tapping improves sharply between ages 3 and 6 (Anthony et al. 2013; Woodward et al. 2016). Touch data alone classifies a child aged 3-6 versus an adult with **86.5% accuracy** (Vatavu et al. 2015). Children aged 2-3 struggled with double tap, long press and pinch-to-shrink (UPV). Sesame Workshop's synthesis of more than 50 internal studies makes tap the most intuitive gesture. It also advises registering taps on touch-down and accepting partial tracing, because children lift their fingers (Cooney Center). Android's 48dp and WCAG's 24px minimums are adult baselines (Android; W3C 2.5.8). No verified source gives a millimetre figure for children. NUHA's 72dp should therefore be the floor for secondary controls only. Primary task targets for ages 2-5 should run well above it, at roughly 96-120dp with 16-24dp gaps; this is an inference to test. The layout experiment already shows the room is there. Ten number tiles fit at **187dp each** on a 1280×800dp landscape screen with no scrolling, and still fit at 188dp with text at 1.5×.

NUHA's own emulator work produced the single most actionable rule. When placed items reflowed into the gap left by a removed card, the next tap hit the wrong object. That is a motor error recorded as a conceptual one. **Targets must never move during a task.** Placed items leave an empty slot. This matches the W3C guidance pattern "Ensure Controls and Content Do Not Move Unexpectedly" (W3C COGA) and the AAC practice of fixed symbol positions for motor planning (AssistiveWare). The same experiments found that text which grows with font scale must sit outside the area used to size targets, and that hand-drawn tracing guides need professionally made vector letterforms rather than system-font text.

Input should be tap-first, with drag offered and never required. Children under about five often lose contact mid-drag. Tap-item-then-tap-destination is the fallback (uxdesign.cc), and NUHA's selection experiment already uses it. W3C dragging guidance supports non-drag alternatives. The stylus evidence is mixed. In one study with 28 children, drag-and-drop had 9% higher throughput with a stylus and an error rate of 9.3% versus 12.6% for the finger (Cassidy et al. 2019). A preschool handwriting study, seen only via a repost, favoured the finger (Research Connections). NUHA should accept both inputs and not require the stylus for under-fives. Sesame also found that preschoolers rest their wrists on the bottom edge and corners and exit apps by accident (The Register). NUHA's fixed bottom action bar can stay, but its child buttons should sit inset from the edge and away from the corners, and exit and settings should not be reachable by children at all. On a horizontal table, the "bottom edge" is wherever the child's forearms rest. That makes the inset more important there, not less.

Layout should keep the existing three bands: instruction strip, task area, and fixed action bar, with no scrolling. Child mode should go no deeper than two levels (home, then activity). Use 2-4 choices per task for the youngest children and up to about 4-6 for older ones; these counts are design judgement, not a sourced figure. Icons in child mode should be concrete pictograms paired with tap-to-hear Arabic audio and a short text label for adults. Abstract symbols such as a gear or hamburger menu should not appear. The adult dashboard should be a visibly different, conventional Material 3 interface behind a gate, with 48dp targets, a navigation rail, and text-first progress reports. Apple requires parental gates in Kids-category apps (Apple). Google Play's Families policy is about data, ads and SDKs (Google Play). Simple maths gates are weak against children aged 8 and over (Art & Logic). If the 4-10 range is chosen, NUHA should use a PIN or a press-and-hold combined with a multi-step task, plus Android's Lock Task (kiosk) mode on a dedicated table. Whatever the gate, the child must always be able to pause or stop without passing it.

Motion, sound and rewards should be brief, informative, and set per child

The fixed rules come from WCAG. Content must not flash more than three times per second. Auto-moving content that lasts more than five seconds needs a way to pause it. At AAA level, users should be able to disable animation triggered by interaction, because of vestibular sensitivity (W3C 2.3.1; W3C 2.2.2; W3C 2.3.3). Android's "Remove animations" setting sets the animator duration scale to 0, and Compose animations respect it. Custom animations and basicMarquee reportedly do not (Inclusive Android Apps). Caregivers rarely know this setting exists, so NUHA needs its own motion setting that defaults to the system value. On task screens, nothing should move idly. Avoid parallax, zoom and spin. Every animation should point at the learning target, for example a glow on the correct tile or a ghost hand tracing the stroke. Feedback animations should be identical every time, not randomized surprises. One weak pilot suggests autistic children prefer repetitive motion over random motion (summary).

Sound needs the most care. Hyperacusis affects **roughly 18-40% of autistic children** (J Neural Transm 2023). A preregistered meta-analysis of 13 randomized studies found that white or pink noise slightly helped participants with ADHD (**g = +0.25**) and hurt those without ADHD (**g = −0.21**) (Nigg et al. 2024). So no single default for background audio can be right. In a resource room shared by several children, sound that helps one child can harm the child next to them. NUHA should ship with Arabic voice prompts on, delivered by a calm, warm, moderate-paced voice. Background music should be off, which also respects families who avoid music. Short, soft-attack, volume-normalized effects should be on. Voice, effects and music should each have their own control, as in the PBS KIDS game Railway Hero (Parenting Special Needs). An optional "focus sound" can be offered per child, never globally. Any audio under speech should be at least 20 dB quieter than the speech, or off (W3C 1.4.7). The incorrect-answer sound should never be a buzzer.

Feedback design can lean on autism-support practice. NCAEP lists prompting, reinforcement, time delay and visual supports among its evidence-based practices (NCAEP). In single-case studies, most-to-least prompting produced fewer errors, and most-to-least prompting with a delay was nearly as fast as least-to-most prompting with fewer errors (Libby et al. 2008). In the interface, that becomes a four-step prompt ladder for every activity. First the guide demonstrates fully. Then the target or the stroke's start dot is highlighted. Then a subtle pulse appears. Finally the child works independently. A configurable delay of about 3-5 seconds comes before each hint, and prompts fade as the child succeeds. A wrong answer is never rejected with a red X. The item gently returns, the next prompt level appears, and the trial is logged as "correct with help." Feedback must arrive within about half a second, because children with ADHD find experienced waiting aversive (SAGE 2020). Feedback should also say what was right, for example "٣, بعد ٢، أحسنت" rather than just "ممتاز."

Rewards cut both ways. A meta-analysis of 128 experiments found that expected, tangible, performance-contingent rewards undermined intrinsic motivation, while informational verbal praise enhanced it (Deci, Koestner & Ryan 1999). That finding is contested (Cameron et al. 2001). In autism and intellectual disability, token economies remain "one of the most important technologies" of applied behaviour analysis (Matson & Boisjoli 2009). The synthesis is a **token board, not a points economy**. A strip of 3-5 star slots fills with each correct or helped-correct trial. A full strip earns a short, calm, child-chosen break. Each trial gets a gentle celebration of 1.5 seconds or less: a soft chime, a small sparkle at the target, and a nod from the guide. Leaderboards, lives, streak penalties, punitive timers and unlock shops should be left out. Visual activity schedules are an evidence-based practice from preschool to adulthood (Knight et al. 2015). This supports a right-to-left session strip, a "أولا… ثم…" (first… then…) card, a fixed trial count shown as dots, and a consistent closing ritual ("انتهينا").

Comparable apps split into "playful" and "predictable," and NUHA needs both

The leading mainstream apps share a recognizable pattern. They use animal or monster mascots, as Khan Academy Kids does with Kodi the Bear and a cast assigned to each subject (App Store). They navigate by voice and pictures. Sago Mini's Jinja's Garden uses no words at all, and won the 2026 Apple Design Award for Interaction after testing with more than 100 children. It replaced a joystick that confused toddlers with tap-to-move (gori.me). These apps avoid timers and game-over screens (Screenwise), carry no ads, and keep parents behind a gate (Common Sense). Special-needs apps put control first instead. Proloquo2Go offers 23 grid sizes and adjustable spacing and contrast (App Store). Choiceworks offers now/next boards with the family's own photos and audio (App Store). Otsimo uses a caregiver screening and an ad-free design "to reduce distractions" (UB HAPPE). NUHA's design should combine the two families. The child layer follows the mainstream pattern: voice-led, icon-first, a mascot, and no failure states. The adult layer follows the special-needs pattern: a profile per child, separate audio channels, motion and contrast controls, and schedules.

Two cautions come from this comparison. Independent efficacy evidence is scarce. A small randomized trial of Khan Academy Kids with about 50 low-income preschoolers raised early-literacy percentiles from 34 to 47 (UMass). The Arabic Feed the Monster evaluation called its evidence "weak but encouraging" (INEE). The apps with the boldest claims have the least independent evidence behind them. Arabic apps such as Lamsa and Abjadiyat stand out on curriculum and language, not on documented design (App Store). No public source describes their typefaces, tashkeel policy, digit form or character dress, so a hands-on audit is still needed.

Recommended style direction

Area	Recommendation	Evidence level
Palette approach	Keep low-saturation teal/sand on warm off-white for large surfaces. Use dark, desaturated teal or warm near-black for text (aim for 7:1). Give every tappable tile a border of at least 3:1, because current container fills are 1.15-1.22:1. Use saturated accents (yellow stars) only in small areas. Never use hue alone for meaning: add shape, icon, border and position. Provide three themes: calm (default), high-contrast, and extra-muted.	Guideline (WCAG, GOV.UK) plus Moderate (saturation and gaze; reduced chromatic discrimination). No evidence that the palette improves learning.
Type	Bundle Noto Naskh Arabic for all vowelized learning content. Use a rounded face (Baloo Bhaijaan 2, or alternatively Readex Pro) for headings and UI after a harakat test. Start learning text at about 28-32sp and UI labels at about 20-24sp. Set line height to 1.6-1.8× for vowelized text. Use no letter-spacing and no tatweel.	Moderate (simplified Naskh for children); Guideline (alreq); sizes are Inference.
Diacritics	Full, native-speaker-checked tashkeel on every learning item at all ages. Light tashkeel on UI labels.	Strong (Abu-Rabia; Saiegh-Haddad).
Numerals	One app-wide setting; never mix systems on a screen; format explicitly. Choose the default only after checking current MoE Grade 1 maths books.	Guideline (W3C); default unresolved.
Illustration	Plain, consistent task backgrounds. Semi-realistic Saudi objects for matching and vocabulary, with photos as an option for ages 2-4. A cartoon-to-photo ladder for emotions, with diverse Saudi faces. Flat style only for UI chrome.	Strong (seductive details, decoration); Moderate (photo transfer in toddlers; cartoon faces in autism).
Character	One calm guide (animal or neutral), present at instruction and feedback, demonstrating tasks, still during work. Adults can hide it or mute its voice.	Moderate (agents g ≈ 0.20; Transporters); Inference for this population.
Components and layout	Three fixed bands; no scrolling; targets never move (placed items leave slots); child buttons inset from edges and corners; 2-4 choices for under-fives; navigation two levels deep; concrete pictogram plus audio label plus text.	NUHA experiment plus Guideline (COGA, Sesame); counts are Inference.
Targets and input	72dp floor; primary targets 96-120dp+ for ages 2-5 with 16-24dp gaps; sizes calculated from available space. Tap fires on touch-down, with debounce. Drag is optional with a tap-tap fallback and snap zones. No double-tap, long-press, pinch or multi-finger gestures in child mode. Stylus and finger both accepted; partial strokes tolerated.	Strong (children miss small targets; gesture difficulty under 5); sizes are Inference.
Motion	No flashing; no idle motion on task screens; short, consistent transitions with no parallax, zoom or spin; feedback motion of 1.5 s or less that points at the target. In-app setting: full, reduced, none (default follows OS).	Guideline (WCAG 2.3.x, 2.2.2) plus Strong (congruent vs distracting multimedia).
Sound	Arabic voice on (Fusha text, familiar vocabulary, optional dialect prompts for under-fives). Music off by default. Soft-attack, normalized effects. No buzzer. Separate voice, effects and music controls. Optional focus noise per child.	Moderate/Strong (hyperacusis prevalence; noise meta-analysis); Market evidence for music-off.

Area	Recommendation	Evidence level
Feedback and rewards	Feedback within about 0.5 s; four-level prompt ladder with a 3-5 s configurable delay and fading; neutral return on error; informative Arabic praise; 3-5 slot token board leading to a calm break; no leaderboards, lives, punitive timers or shops.	Guideline/Moderate (NCAEP; single-case prompting studies); Strong but contested (overjustification); Inference on intensity.
Progress	Right-to-left session schedule strip; first/then card; fixed trial dots; a closing ritual.	Strong-ish (visual schedules systematic review).
Adult UI	Separate Material 3 theme (48dp, navigation rail, text-first) behind a PIN or hold-plus-task gate. Lock Task mode on the table. Per-child sensory profile. The child can always pause.	Guideline (Apple, Play, Sesame); Inference on design.
Settings per child	Theme, motion level, three audio channels, focus sound, celebration intensity (off, gentle, lively), character on/off, prompt delay, target-size step, numeral system.	Strong rationale (sensory subtypes; ADHD noise split) plus Guideline (COGA Objective 8).

What must be tested with children on the real table

Desk research cannot settle the following, and each should be a task in NUHA's formative sessions. The first is **target size and spacing by age**: miss rates at 72, 96 and 120dp on the chosen table, including reach to far targets and accidental palm and wrist contacts at the near edge. The second is **palette acceptance across sensory profiles**: whether hyper-responsive children tolerate the calm theme, whether sensory-seeking children disengage, and whether the bordered tiles are found faster than unbordered ones. The third is **Arabic font legibility**: Noto Naskh against a rounded candidate, at two sizes, with fully vowelized words, measured by children's letter and word identification rather than adult preference. The fourth is **guide character presence**: on, off, and instruction-only, comparing task completion and off-task looking. The fifth is **feedback and celebration intensity**, comparing gentle and lively styles for children with ADHD and autism, and the effect of the neutral error response on distress and retries. The sixth is **input modes**: tap-tap against drag for matching, and finger against stylus for tracing, with under-fives tested separately. The seventh is **icon and arrow comprehension in RTL**, meaning whether pre-readers read the mirrored "next" and the schedule strip correctly. The eighth is **voice register**: Fusha against a Saudi dialect prompt for the youngest children. The last is **adult interpretation** of the dashboard and **gate bypass** by older children. The age-range decision changes several defaults, including target size, number of choices, gate strength, the photo-versus-illustration choice, and how much text appears. That makes it the first owner decision to close.

How it maps to Jetpack Compose

Most of this direction maps cleanly onto the existing NuhaTheme. The API names below come from the research notes and author knowledge; check them against current documentation before relying on them.

Requirement	Compose approach
RTL always	Keep CompositionLocalProvider(LocalLayoutDirection provides LayoutDirection.Rtl) at the root, as now. Use Icons.AutoMirrored and android:autoMirrored for directional icons. Wrap tracing canvases, media controls and charts in a LayoutDirection.Ltr subtree where coordinates must not flip.

Requirement	Compose approach
Two modes	Two themes: NuhaChildTheme (custom shapes, large type, bordered tile tokens) and NuhaAdultTheme (standard M3 scales, navigation rail via currentWindowAdaptiveInfo()). Provide LocalMinimumInteractiveComponentSize as 72.dp or larger in child mode and 48.dp in adult mode.
Fonts	Bundle subsetted OFL fonts in res/font as FontFamily values and assign them in Typography. Raise lineHeight to 1.6-1.8× on display and headline styles used for learning text. Never set letterSpacing on Arabic text. Run render tests for clipping at 1.0× and 1.5× font scale.
Numerals	Format all numbers through one helper using NumberFormat/DecimalFormatSymbols with Locale.forLanguageTag("ar-SA-u-nu-arab") or "-u-nu-latn", driven by the per-child setting.
Contrast tokens	Add explicit tileBorder and tileSelectedBorder tokens (3:1 or more), and add WCAG ratio unit tests for every foreground/background pair in each theme.
Stable targets	Size tiles with BoxWithConstraints from the task area only, as the layout experiment does. Use keyed slots that keep their position when emptied, and never reflow a Row or LazyRow during a trial.
Tap feedback	Use pointerInput { detectTapGestures(onPress = …) } for touch-down highlight with a short debounce. Make each game element a single semantics node with a name, state and custom actions (as already validated).
Drawing	Use Ink InProgressStrokes with relative coordinates (as validated). Draw vector letter guides, not text. Stylus-only and palm rejection still need tests on the real device.
Motion	Create a LocalNuhaMotion (full, reduced, none) seeded from Settings.Global.ANIMATOR_DURATION_SCALE and overridden per child. Route all custom animations through it. Avoid basicMarquee and any infinite transitions on task screens.
Audio	Use separate players or volume buses for voice, effects and music (for example SoundPool for effects and Media3 for voice), each scaled by a setting per child. Normalize loudness of all assets. Keep music off by default.
Settings	Store a per-child SensoryProfile in Room and expose it through a CompositionLocal, so that every component reads theme, motion, audio, celebration level and target-size step from one source.
Kiosk and gate	Use Lock Task mode on the dedicated table. Put the adult area behind a PIN or a hold-plus-task gate, and keep a pause control the child can always reach.

Conclusion

The research shifts the question from "which colours calm autistic children" to "which defaults do least harm, and who adjusts them." The colour evidence is too thin to defend a single palette on learning grounds. The evidence on structure is not thin: sparse scenes, congruent multimedia, stable targets, prompting with fading, and visual schedules all have support. Neither is the evidence on individual differences, where sensory subtypes and the ADHD noise split mean that one setting helps some children and harms others. NUHA's real design asset is therefore its settings model as much as its look. A calm, bordered, Naskh-and-tashkeel child interface becomes the starting point, and therapists in resource rooms raise or lower the stimulation for each child from there.

Two facts specific to NUHA stand out. The mis-tap finding from its own emulator experiment is stronger evidence for "never move targets" than anything in the literature, because it shows a motor slip being recorded as a learning error. That turns layout stability into a data-integrity rule, not only a usability rule. And the current theme already passes text contrast, but its tinted tiles are essentially invisible as controls by WCAG's measure. That is a cheap fix to make before any child testing starts. The decisions still open are the age range, the numeral convention and the voice register. They are owner and curriculum decisions, and closing them would unlock most of the remaining defaults.